1 **(a)** $\frac{\sqrt{3}}{2}$ **[B1]**

(b) $\sin 30° = \frac{1}{2}$ and $\tan 60° = \sqrt{3}$ **[M1]**

$\frac{1 - 1/2}{(\sqrt{3})^2} = \frac{1/2}{3} = \frac{1}{6}$ with no errors seen **[A1]**

[Total Marks: 3]

2 **(a)** 2 **[B1]**

(b) $\frac{1}{2}$ **[B1]**

(c) $\frac{\sqrt{3}}{\sqrt{3}/2}$ **[M1]**

2 **[A1]**

[Total Marks: 4]

3 **(a)** Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ and $\tan 30° = \frac{1}{\sqrt{3}}$ (or $\frac{\sqrt{3}}{3}$) **[M1]**

1 **[A1]**

(b) Substitutes $\tan 45° = 1$ leading to $\cos x = \frac{1}{\sqrt{2}}$ **[M1]**

45 **[A1]**

[Total Marks: 4]

4 **(a)** $\frac{1}{2} + \frac{1}{2} = 1$ **[B1]**

(b) States at least two exact values correctly: $\tan 60° = \sqrt{3}$, $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 0° = 1$ **[B1]**

Substitutes into expression: $\sqrt{3} \times \frac{\sqrt{3}}{2} - 1 = \frac{3}{2} - 1$ **[M1]**

$\frac{1}{2}$ (or 0.5) **[A1]**

[Total Marks: 4]

5 **(a)** **(i)** $\sin 30° = \frac{1}{2}$ and $\cos 60° = \frac{1}{2}$ **[M1]**

1 **[A1]**

(ii) $\tan 60° = \sqrt{3}$ and $\cos 30° = \frac{\sqrt{3}}{2}$, giving $\frac{\sqrt{3}}{\frac{\sqrt{3}}{2}} = 2$ **[B1]**

$(\sin 45°)^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$ **[B1]**

$\frac{3}{2}$ (or 1.5 or $1\frac{1}{2}$) **[A1]**

(b) $\cos x = \frac{\sqrt{3}}{2}$ **[M1]**

30 **[A1]**

(c) $\tan y = 1 - 1 = 0$ **[M1]**

0 **[A1]**

[Total Marks: 9]

6 **(a)** $\sin 45° = \frac{6}{PR}$ (or $\cos 45° = \frac{6}{PR}$) **[M1]**

$PR = 6 \div \frac{\sqrt{2}}{2} = 6\sqrt{2}$ **[A1]**

(b) $\sin 60° = \frac{6\sqrt{2}}{PS}$ **[M1]**

$PS = 6\sqrt{2} \div \frac{\sqrt{3}}{2}$ **[M1]**

$PS = \frac{12\sqrt{2}}{\sqrt{3}} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}$ **[A1]**

[Total Marks: 5]

7 **(a)** $\sin 45° = \frac{PQ}{8\sqrt{2}}$ **[M1]**

$PQ = 8\sqrt{2} \times \frac{1}{\sqrt{2}} = 8$ **[A1]**

(b) $\cos 30° = \frac{8\sqrt{2}}{PS}$ or $PS = \frac{8\sqrt{2}}{\cos 30°}$ **[M1]**

$PS = \frac{8\sqrt{2}}{\sqrt{3}/2} = \frac{16\sqrt{2}}{\sqrt{3}}$ **[M1]**

Rationalises the denominator to get $\frac{16\sqrt{6}}{3}$ **[A1]**

[Total Marks: 5]

8 In $\triangle ABD$, $AD = 8\sqrt{2}\cos 45° = 8$ **[M1]**

$BD = 8\sqrt{2}\sin 45° = 8$ **[A1]**

In $\triangle BDC$, $BC = \frac{BD}{\sin 30°} = \frac{8}{0.5} = 16$ **[M1]**

$DC = \frac{BD}{\tan 30°} = 8\sqrt{3}$ **[M1]**

Perimeter $= AB + BC + AD + DC = 8\sqrt{2} + 16 + 8 + 8\sqrt{3}$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Total Marks: 6]

9 **(a)** Uses $\sin 45° = \dfrac{BD}{8}$ with $\sin 45° = \dfrac{\sqrt{2}}{2}$ (or $\dfrac{1}{\sqrt{2}}$) **[M1]**

$4\sqrt{2}$ **[A1]**

(b) Uses $\sin 30° = \dfrac{4\sqrt{2}}{BC}$ with $\sin 30° = \dfrac{1}{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) Finds $AD = 8\cos 45° = 4\sqrt{2}$ and $DC = \dfrac{4\sqrt{2}}{\tan 30°}$ (or $8\sqrt{2}\cos 30°$) **[M1]**

Evaluates $DC = 4\sqrt{2} \times \sqrt{3} = 4\sqrt{6}$ and sums $AC = AD + DC = 4\sqrt{2} + 4\sqrt{6}$ **[A1]**

[Total Marks: 6]

10 **(a)** $BD = 8\sqrt{2}\sin 45° = 8\sqrt{2} \times \dfrac{\sqrt{2}}{2} = 8$ cm **[M1]**

$AD = 8\sqrt{2}\cos 45° = 8\sqrt{2} \times \dfrac{\sqrt{2}}{2} = 8$ cm **[A1]**

(b) $\tan 30° = \dfrac{8}{CD}$ giving $\dfrac{1}{\sqrt{3}} = \dfrac{8}{CD}$ **[M1]**

$8\sqrt{3}$ **[A1]**

(c) Base $AC = 8 + 8\sqrt{3}$ cm **[B1]**

Area $= \dfrac{1}{2}(8 + 8\sqrt{3}) \times 8$ **[M1]**

$32 + 32\sqrt{3}$ **[A1]**

(d) $\sin 30° = \dfrac{8}{BC}$ giving $BC = \dfrac{8}{\frac{1}{2}} = 16$ cm **[M1]**

Perimeter $= 8\sqrt{2} + 16 + (8 + 8\sqrt{3})$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Total Marks: 10]

11 **(a)** Substitutes $\tan 30° = \dfrac{1}{\sqrt{3}}$ to get $\tan^2 30° = \dfrac{1}{3}$ **[B1]**

Evaluates numerator as $1 - \dfrac{1}{3} = \dfrac{2}{3}$ AND denominator as $1 + \dfrac{1}{3} = \dfrac{4}{3}$ **[M1]**

Simplifies $\dfrac{2/3}{4/3} = \dfrac{1}{2}$ which is equal to $\cos 60°$ **[A1]**

(b) Substitutes exact values: $x\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} - x\left(\frac{1}{2}\right)$ **[M1]**

Rearranges to collect x terms: $x\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \sqrt{3}$ **[M1]**

Isolates x: $x = \frac{2\sqrt{3}}{\sqrt{3}+1}$ **[M1]**

Rationalises denominator: $\frac{2\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{6-2\sqrt{3}}{2} = 3 - \sqrt{3}$ **[A1]**

[Total Marks: 7]

12 (a) $\cos 30° = \frac{AD}{12}$ **[M1]**

$AD = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3}$ with no errors seen **[A1]**

(b) Uses Area $= AB \times AD = 60\sqrt{3}$ to find $AB = 10$ **[B1]**

$DE = 12\sin 30° = 6$ **[M1]**

$CE = CD - DE = 10 - 6 = 4$ **[A1]**

(c) Identifies $BC = 6\sqrt{3}$ and $CE = 4$ **[M1]**

$\tan(\angle EBC) = \frac{CE}{BC} = \frac{4}{6\sqrt{3}}$ **[M1]**

$\frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9}$ **[A1]**

[Total Marks: 8]

13 (a) Uses $h = 4\sqrt{3}\sin 60°$ with $\sin 60° = \frac{\sqrt{3}}{2}$ **[M1]**

6 **[A1]**

(b) Finds left base segment: $4\sqrt{3}\cos 60° = 4\sqrt{3} \times \frac{1}{2} = 2\sqrt{3}$ **[B1]**

Finds right base segment: $\frac{6}{\tan 45°} = 6$, giving total base $SR = 2\sqrt{3} + 6 + 6 = 12 + 2\sqrt{3}$ **[B1]**

Substitutes into trapezium area formula: $\frac{1}{2}(6 + 12 + 2\sqrt{3}) \times 6$ **[M1]**

$54 + 6\sqrt{3}$ **[A1]**

(c) Finds slanted side QR: $\frac{6}{\sin 45°} = \frac{6}{\frac{\sqrt{2}}{2}} = 6\sqrt{2}$ **[M1]**

Sums all four sides: $6 + 6\sqrt{2} + (12 + 2\sqrt{3}) + 4\sqrt{3} = 18 + 6\sqrt{3} + 6\sqrt{2}$ **[A1]**

[Total Marks: 8]

14 **(a)** Uses cosine rule: $7^2 = x^2 + (x+3)^2 - 2(x)(x+3)\cos 60°$ **[M1]**

Substitutes $\cos 60° = 0.5$: $49 = x^2 + x^2 + 6x + 9 - x(x+3)$ **[M1]**

Expands and simplifies: $49 = 2x^2 + 6x + 9 - x^2 - 3x$ **[M1]**

$x^2 + 3x - 40 = 0$ **[A1]**

(b) $(x+8)(x-5) = 0$ **[M1]**

$x = 5$ (rejects $x = -8$) **[A1]**

Perimeter $= 5 + (5+3) + 7 = 20$ **[A1]**

(c) Uses sine area formula: Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Total Marks: 10]

15 **(a)** Angle $OAP = 90°$ (tangent to radius $= 90°$) and angle $APO = 30°$ **[B1]**

$\tan 30° = \frac{6}{PA}$ giving $\frac{1}{\sqrt{3}} = \frac{6}{PA}$ **[M1]**

$6\sqrt{3}$ **[A1]**

(b) Area of quadrilateral $OAPB = 2 \times \frac{1}{2} \times 6 \times 6\sqrt{3} = 36\sqrt{3}\ cm^2$ **[M1]**

Angle $AOB = 360° - 90° - 90° - 60° = 120°$ **[B1]**

Area of minor sector $OAB = \frac{120}{360} \times \pi \times 6^2 = 12\pi\ cm^2$ **[M1]**

$36\sqrt{3} - 12\pi$ **[A1]**

(c) **(i)** Half-chord $= 6\sin 60° = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$ cm **[M1]**

$6\sqrt{3}$ **[A1]**

(ii) $OP = \frac{6}{\sin 30°} = 12$ cm and distance from O to $AB = 6\cos 60° = 3$ cm **[M1]**

$12 - 3 = 9$ **[A1]**

[Total Marks: 11]